

The Geometry of Linear Equations

One question, three ways to see it: the **row picture**, the **column picture** and the **matrix form** $Ax = b$.

Explained from zero on a real money question: splitting ₹10 lakh between two funds.

2 × 2 and 3 × 3

Portfolio mix

MIT 18.06, Lecture 1

THE QUESTION

You have ₹10 lakh and two funds

Fund	Return per year	On ₹1 lakh
Nifty 50 index fund	12%	₹12 thousand
Government bond fund	7%	₹7 thousand

You want to invest all ₹10 lakh and earn ₹1,05,000 a year.

THE QUESTION

How much goes into each fund?

Returns are illustrative round numbers, not forecasts.

STEP 1

Turn it into equations

Let $x = ₹ \text{ lakh in Nifty}$ and $y = ₹ \text{ lakh in bonds}$. Each requirement is one equation:

$$x + y = 10 \quad \text{budget (₹ lakh)}$$

$$12x + 7y = 105 \quad \text{return (₹ thousand)}$$

Two equations that must both hold are a **system**. Each unknown is only multiplied by a number and added, so the equations are **linear**.

THE FUNDAMENTAL PROBLEM OF LINEAR ALGEBRA

Solve n linear equations in n unknowns.

Three ways to read the same equations

View	You look at	The question becomes
Row picture	one equation at a time	Where do the lines meet ?
Column picture ★	one fund at a time	Which mix of fund arrows hits the target?
Matrix form	everything at once	Find x with $Ax = b$.

The ★ marks the view the whole course is built on.

STEP 2

Each fund is a vector

₹1 lakh of a fund uses ₹1 lakh of budget and earns its return. Stack the two numbers in a column:

$$\text{Nifty} = \begin{bmatrix} 1 \\ 12 \end{bmatrix} \quad \text{Bond} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

- **Scale:** $7(1, 12) = (7, 84)$: ₹7 lakh of Nifty.
- **Add:** $(1, 12) + (1, 7) = (2, 19)$: ₹1 lakh in each.

LINEAR COMBINATION

Scale vectors and add them: $x c + y d$. Here, a linear combination of the funds is simply a **portfolio**.

STEP 3

Matrix form: $Ax = b$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 10 \\ 105 \end{bmatrix}}_b$$

Rows of A = the rules (budget, return). **Columns of A** = the funds.

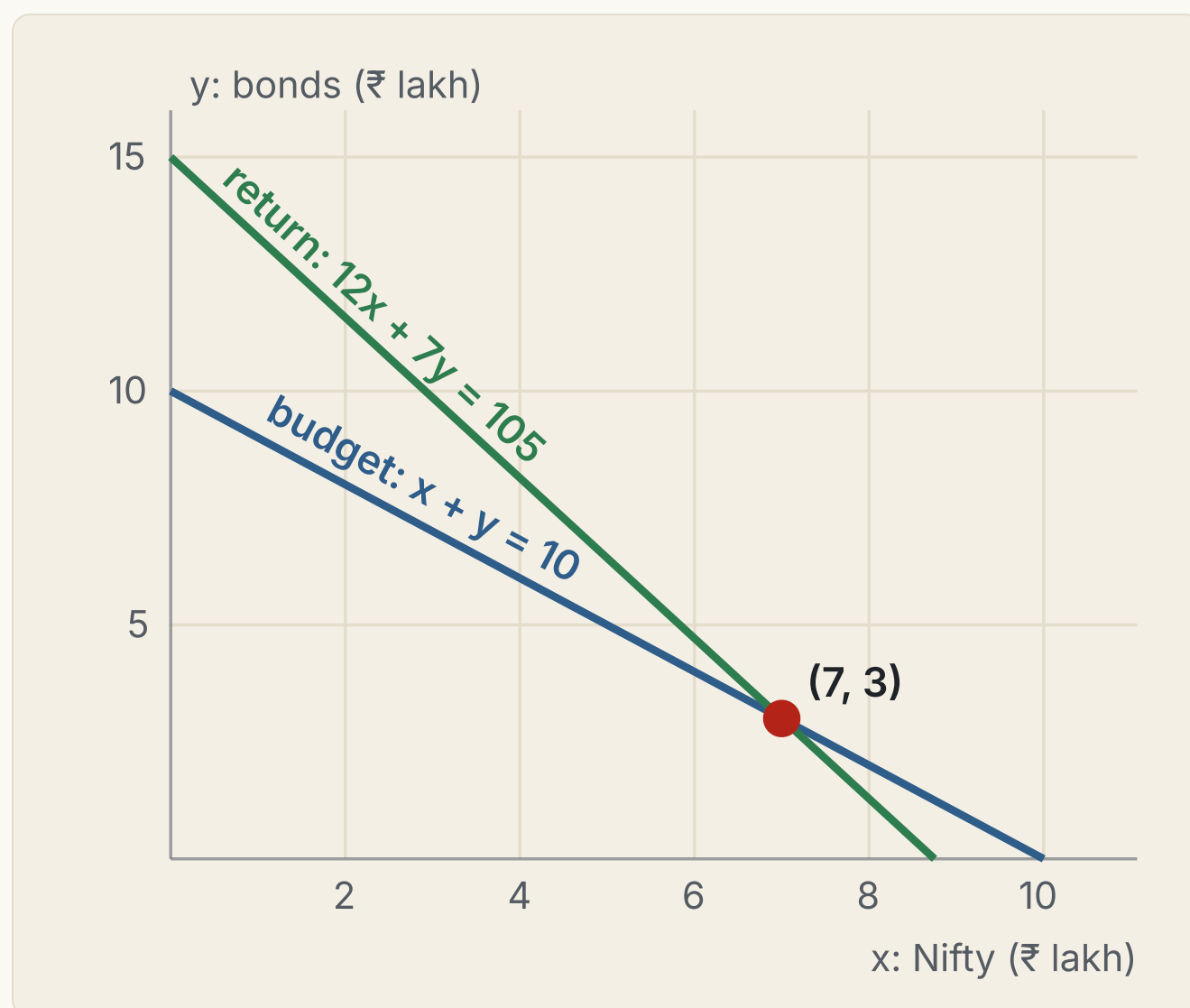
By columns: $A \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}$

By rows: $1 \cdot 7 + 1 \cdot 3 = 10, \quad 12 \cdot 7 + 7 \cdot 3 = 105$

KEY FACT

Ax is a combination of the columns of A .

Row picture: lines that cross

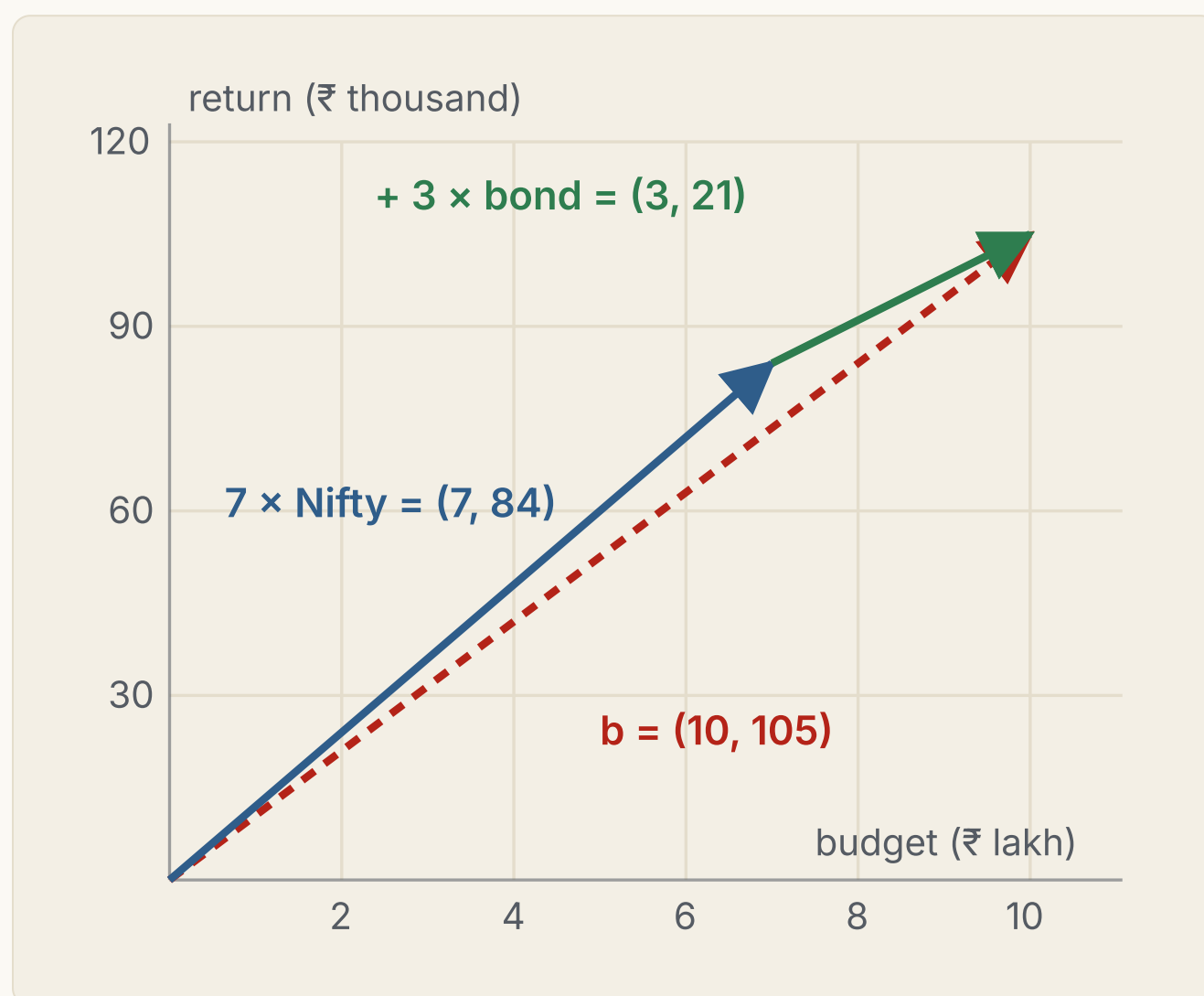


Each rule is a line. The answer is where they cross:
₹7 lakh Nifty, ₹3 lakh bonds.

Check: $7 + 3 = 10$ ✓ and $84 + 21 = 105$ ✓

Column picture: mix the arrows

$$x \begin{bmatrix} 1 \\ 12 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}$$



Same answer, seen as a **mix of fund arrows** landing exactly on the target b .

Which targets can we reach?

The **span** of some vectors = every point you can reach by mixing them.

Funds you may use	Span	Meaning
Nifty only	a line	always exactly 12%
Bonds only	another line	always exactly 7%
Nifty + bonds	the whole plane	any budget, any return

COLUMN SPACE $C(A)$

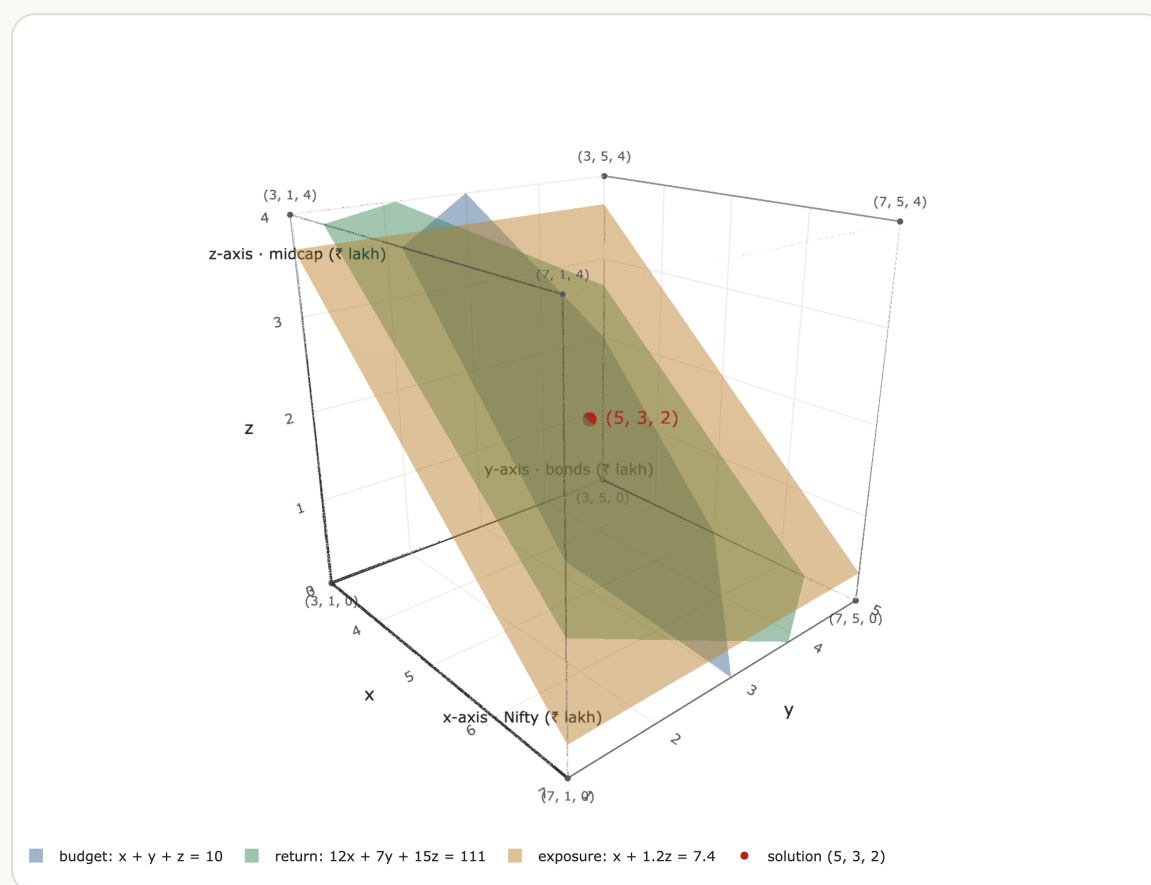
The span of the columns of A : every target your funds can produce.

WHY IT MATTERS

$Ax = b$ has a solution exactly when b is in $C(A)$.

Add a midcap fund and a third rule

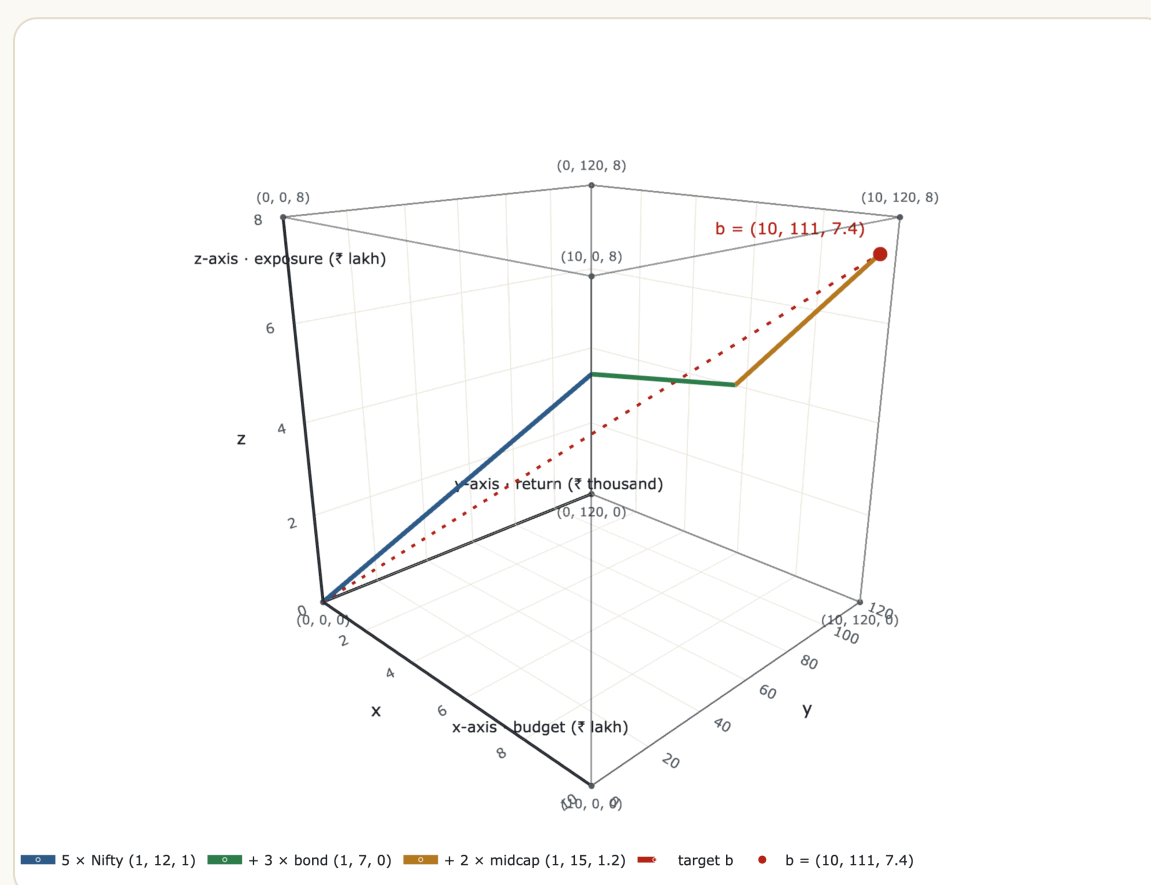
$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix} \quad \begin{array}{l} \text{budget} \\ \text{return} \\ \text{market exposure} \end{array}$$



Row picture: three planes meet at one point: ₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap.

Column picture in 3D

$$5 \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 15 \\ 1.2 \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix}$$



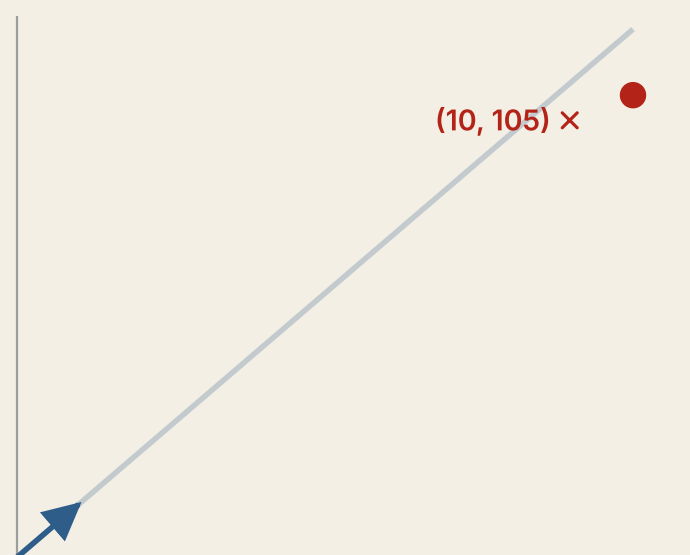
Change the target and the row picture gets three new planes. The column picture keeps the same three arrows, just mixed differently.

Singular matrices

Two Nifty funds (both 12%) point the same way:



Parallel lines: no solution.



Every mix earns 12%: the target is off the line.

SINGULAR VS INVERTIBLE

Invertible: the columns fill the space: one solution for every b . **Singular:** a fund adds no new direction (a twin fund, or a 50/50 balanced fund): no solution or infinitely many.

Key takeaways

1. A linear system is $Ax = b$: rows are rules, columns are funds.
2. **Row picture**: lines (or planes) meet at the solution.
3. **Column picture ★**: b is a mix of the columns; the solution gives the amounts.
4. Ax is a combination of the columns of A .
5. Every target is reachable exactly when the columns fill the space (A invertible).
6. A fund built from others adds no direction: A is singular.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

Fund mixer, 3D plots you can rotate, worked examples and practice:

learn.krisarj.com

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